Sheeraz Athar

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Research Interests

Tactile sensor design and fabrication • Dexterous robotic hand and mechanism design • Soft and bio-inspired robotics • Contact-rich manipulation • Imitation learning and teleoperation • Embodied AI • Sim-to-real transfer • Multimodal sensing for surgical and underwater robotics

Education

Purdue University — *West Lafayette, IN, USA*
Ph.D., Industrial Engineering — *2022 – 2026*

- Dissertation: *Multimodal Tactile Sensing for Contact-Rich Robotic Manipulation*
- Advisor: Prof. Yu She (MARS Lab)
- Committee: Prof. Qiang Qiu (Purdue ECE), Prof. Wenzhuo Wu (Purdue IE), Prof. Laura Blumenschein (Purdue ME)

Hong Kong University of Science and Technology (HKUST) — *Hong Kong*
M.Phil., Mechanical Engineering — *2019 – 2021*

- Thesis: *3D Printable Origami-Inspired Soft Actuators for Robotic Applications*
- Advisor: Prof. Michael Yu Wang, Cheng Kar-Shun Robotics Institute

Aligarh Muslim University (AMU) — *Aligarh, India*
B.Tech., Mechanical Engineering — *2015 – 2019*

- Thesis: *EEG-Based Exoskeleton for Finger Rehabilitation*
- Advisor: Prof. M. Muzammil

Academic Appointments

University of Notre Dame — *Notre Dame, IN, USA*
Postdoctoral Researcher, Physical AI & Robotics (PAIR) Initiative — *Sep 2026 – Present*

- Advisor: Prof. Nikolaus Correll
- Developing dexterous robotic hands with integrated tactile sensing through hardware–software co-design, including sim-to-real morphology optimization using MuJoCo and the SR^2 self-reconfigurable hand platform.
- Developing imitation learning and teleoperation pipelines for contact-rich manipulation using the Unitree G-1 humanoid, Magpie hand, Dex-3 hand, and UR16e robot arm.
- Investigating sim-to-real transfer of dexterous manipulation policies, with a focus on bridging simulation-to-reality gaps in hand morphology and contact dynamics.

Purdue University — *West Lafayette, IN, USA*
Graduate Research Assistant, MARS Lab — Advisor: Prof. Yu She — *2022 – 2026*

- Designed and deployed four multimodal tactile sensing platforms integrating vision, vibration, acoustic, thermal, and photoelastic sensing modalities, including soft and bio-inspired designs such as *FibTac* a fiber-embedded pneumatic gripper with embodied tactile sensing.
- Designed the VTAP Gripper — a dexterous end-effector combining fingertip sensing with a visuo-tactile active palm for in-hand manipulation (**IROS 2026 Best Paper Award Finalist**).
- Collaborated with Prof. Jun Ueda and Prof. Ye Zhao (Georgia Tech) on multimodal perception and robot learning for contact-rich manipulation.
- Collaborated with Prof. Qiang Qiu (Purdue ECE) on multimodal learning and deep perception algorithms for tactile sensing.
- Contributed to NSF Foundational Research in Robotics (FRR) proposal development and USDA proposal preparation under Prof. Yu She.

Hong Kong University of Science and Technology (HKUST) *Hong Kong*
Graduate Research Assistant, Robotics Institute — Advisor: Prof. Michael Yu Wang *2019 – 2021*

- Designed origami-inspired rotary and bi-directional soft actuators for grippers and quadruped robots; published at IEEE IROS 2020.
- Developed a 3D printing method for Kevlar-reinforced foldable soft robotic structures.

École de Technologie Supérieure (ÉTS) *Montréal, Canada*
Research Intern, Biomechanics Group — Advisor: Prof. Eric Wagnac *Jun – Aug 2018*

- Developed finite element models of the human spine (T11–L1) to evaluate ligament injury risk under flexion/extension and trauma loading.

Publications

Refereed Journal Articles

[1] **S. Athar**, X. Zhang, M. R. I. Prince, V. G. Duffy, and Y. She, "*FibTac*: A Fiber-Based Pneumatic Gripper with Embodied Tactile Sensing," *npj Robotics* (Nature Portfolio), vol. 4, Art. no. 47, Aug. 2026. doi:10.1038/s44182-026-00112-0 [Code]

[2] **S. Athar**, X. Zhang, J. Ueda, Y. Zhao, and Y. She, "*VibTac*: A High-Resolution, High-Bandwidth Tactile Sensing Finger for Multi-Modal Perception in Robotic Manipulation," *IEEE Transactions on Haptics*, vol. 18, pp. 507–518, 2025. doi:10.1109/TOH.2025.3561049 [Video]

[3] M. R. I. Prince, **S. Athar**, P. Zhou, and Y. She, "*TacScope*: A Miniaturized Vision-Based Tactile Sensor for Surgical Applications," *Advanced Robotics Research*, Wiley, 2025. doi:10.1002/adrr.202500117 [Video]

[4] **S. Athar**, G. Patel, Z. Xu, Q. Qiu, and Y. She, "*VisTac*: Towards a Unified Multimodal Sensing Finger for Robotic Manipulation," *IEEE Sensors Journal*, vol. 23, no. 20, pp. 25440–25450, 2023. doi:10.1109/JSEN.2023.3310918 [Video]

[5] **S. Athar**, C. Hillyer, V. Vivek, S. Shelar, A. Song, N. Nguyen, A. Patel, C. Huang, and Y. She, "Design and Development of a Multimodal Underwater Robot: An All-in-One Drifter, Glider, and Thruster," *ASME Letters in Translational Robotics*, vol. 2, no. 3, Art. 031002, 2026. NSF-funded. doi:10.1115/1.4072705 [Press]

Refereed Conference Papers

[6] Y. Zhou, **S. Athar**, Z. Hu, B. Huang, Y. Li, J. Wachs, and Y. She, "*VTAP Gripper*: Synergizing Fingertip Sensing and a Visuo-Tactile Active Palm for Dexterous In-Hand Manipulation," *IEEE/RSJ IROS*, 2026. **[IROS 2026 Best Paper Award Finalist] [ASME SMRDC 2nd Prize]** [arXiv] [Website]

[7] Z. Hu, Z. Xu, **S. Athar**, J. Wachs, and Y. She, "*MuxGel*: Simultaneous Dual-Modal Visuo-Tactile Sensing via Spatially Multiplexing and Deep Reconstruction," *IEEE/RSJ IROS*, 2026. [Preprint]

[8] M. R. I. Prince, J. Kim, Y. Zhou, M. Vrshek, A. Khera, **S. Athar**, et al., "*MVP-Tac*: A Miniaturized Dual-Modal Vision and Photoelastic Tactile Sensor for Robot-Assisted Minimally Invasive Surgery," *IEEE/RSJ IROS*, 2026. [Website]

[9] S. Liu, **S. Athar**, and M. Y. Wang, "Vacuum Driven Auxetic Switching Structure and Its Application on a Gripper and Quadruped," *IEEE/RSJ IROS*, 2020, pp. 8829–8834. doi:10.1109/IROS45743.2020.9341338

Manuscripts Under Revision

[10] **S. Athar**, Q. Lu, Z. Zhang, M. R. I. Prince, M. Reed, Z. Teschner, D. Bengani, and Y. She, "*TriSense*: A Unified Visual–Tactile–Auditory Robotic Finger for Learning-Enhanced Contact-Rich Manipulation," *Manuscript under revision.*

[11] X. Zhang, **S. Athar**, V. G. Duffy, and Y. She, "*ThermoTac*: High-Resolution Thermal Tactile Sensing for Multi-Phase Robotic Perception," *Manuscript under revision.*

Patent Applications

[1] **S. Athar**, X. Zhang, M. R. I. Prince, V. G. Duffy, and Y. She, *System and Method for Fiber-Based Gripper with Integrated Tactile Sensing.* U.S. Provisional Patent Application, filed 2025.

[2] **S. Athar**, Q. Lu, Z. Zhang, M. R. I. Prince, M. Reed, Z. Teschner, D. Bengani, J. Wachs, and Y. She, *Multimodal In-Finger Sensing and Control for Robotic Manipulation Using Through-Contact Vision, Tactile Imaging, and Vibration Detection.* U.S. Provisional Patent Application, filed 2025.

[3] Y. Zhou, **S. Athar**, Z. Hu, and Y. She, *Tactile-Reactive Robotic Gripper with Actuated Palm and Multi-Modal Sensing.* U.S. Provisional Patent Application, filed 2025.

[4] **S. Athar**, C. Hillyer, V. Vivek, S. Shelar, A. Song, N. Nguyen, A. Patel, C. Huang, and Y. She, *Underwater Robotic Systems Having Multiple Selectable Locomotion Modes.* U.S. Provisional Patent Application, filed 2025.

Grants & Proposal Writing

- **NSF Foundational Research in Robotics (FRR) —** Contributed to proposal preparation including technical narrative drafting and preliminary experimental results. (PI: Prof. Yu She, Purdue University)
- **USDA Agricultural Robotics Proposal —** Assisted with early-stage concept development and preparation of preliminary figures for a robotic grain monitoring system. (PI: Prof. Yu She, Purdue University)

Teaching Experience

Graduate Teaching Assistant *Purdue University, 2022–2025*
School of Industrial Engineering

- **IE 574: Industrial Robotics and Flexible Assembly —** Designed and led laboratory modules on real-time object detection, visual servoing, and multi-robot coordination using CoppeliaSim, TMFlow, Python, and FetchCore. Supervised five student team projects through concept, implementation, and final demonstration. Delivered an invited guest lecture on multimodal tactile sensing (Spring 2024). Selected student project videos: [V1] [V2] [V3] [V4]

- **IE 370: Manufacturing Processes** — Led weekly laboratory sessions covering traditional and advanced manufacturing methods; provided individualized academic support.
- **IE 472: Imagine, Model, and Make** — Taught 3Ds Max for 3D design, rendering, and animation; supervised project-based coursework and evaluated final design deliverables.

Graduate Teaching Assistant *HKUST, Hong Kong, 2020–2021*
Cheng Kar-Shun Robotics Institute

- **Design and Manufacturing** — Taught SolidWorks for mechanical design, assembly modeling, and system-level CAD; guided student teams through a semester-long go-kart assembly project with design reviews and manufacturability feedback.

Research Mentorship

- **Cindy (Sijie) Huang** (Undergraduate, University of Notre Dame) — Purdue SURF Program, Summer 2025
  Research topic: Design and prototyping of a multimodal autonomous underwater vehicle; supervised mechanical subsystem integration, sensor testing, and experimental validation. Outcome: research presented at the 2025 Purdue Summer Undergraduate Research Symposium. [SURF Abstract]
- **Andrew Song** (Undergraduate, Purdue University) — 2024–2026
  Research topic: Mechanical design, fixture development, and data acquisition for tactile sensing systems; contributed to the underwater robot project published in ASME Letters in Translational Robotics (2026).
- **Craig Hillyer** (Undergraduate, Purdue University) — 2023–2026
  Research topic: Fabrication, characterization, and experimental validation of vision-based tactile sensor prototypes; contributed to the underwater robot project published in ASME Letters in Translational Robotics (2026).

Professional Service

Peer Review

- *IEEE Transactions on Robotics* (TRO)
- *IEEE Transactions on Automation Science and Engineering* (TASE)
- *IEEE Robotics and Automation Letters* (RA-L)
- *Soft Robotics* (Mary Ann Liebert)
- *IEEE International Conference on Robotics and Automation* (ICRA)
- *IEEE/RSJ International Conference on Intelligent Robots and Systems* (IROS)

Organizational Service

- **Lab Operations and Procurement**, MARS Lab, Purdue University (2022–2026) — Responsible for vendor coordination, purchasing, and budget operations supporting five concurrent robotics research projects.
- **Seminar Organizer**, Institute of Control, Optimization, and Networks (ICON), Purdue University — Coordinated speaker invitations and logistics for weekly research seminar series.
- **Conference Organizer**, ICON Student Research Conference, Purdue University — Assisted with scheduling, event logistics, and coordination of student presentations.
- **Vice-Chairperson**, ASME Student Chapter, AMU — Organized technical seminars, professional development events, and a university-wide tour to the India International Trade Fair.
- **Professional Memberships**: IEEE Robotics and Automation Society; IEEE Student Member.

Outreach & Diversity

- **Purdue FIRST Robotics** (Sep 2025) — Delivered robotics demonstrations for FIRST Robotics students on mechanical design, sensing, and research programming.
- **MITE — Multi-Ethnic Introduction to Engineering** (Jun–Jul 2024, 2025) — Robotics demonstrations, lab tours, and informal mentoring for pre-college students from underrepresented communities; organized by Purdue's Minority Engineering Program.
- **Blue Sky Summer Scholars** (Jun 2025) — Hosted Ivy Tech Community College students in the MARS Lab for robot demonstrations and a discussion of pathways into engineering research.
- **PREFACE Summer Program** (Jun 2024, 2025) — Mentoring and robotics introduction for incoming first-year engineering students, many of them first-generation college students.
- **Advisor and Mentor**, Autonomous Robotics Club (ARC), Purdue University — Guided students on robot design, perception pipelines, and competition preparation.

Awards & Honors

- **IROS 2026 Best Paper Award Finalist** — VTAP Gripper, IEEE/RSJ IROS, 2026
- **ASME Student Mechanism & Robot Design Competition (SMRDC), 2nd Prize** — VTAP Gripper, ASME IDETC, 2026
- **Postgraduate Studentship**, HKUST, 2019–2021 (competitive fellowship, full financial support)
- **Sir Syed Global Scholar Award**, Sir Syed Society of North America, 2018
- **S.N. Bose Scholars Program**, Government of India & U.S. National Science Foundation, 2018
- **Mitacs Globalink Research Internship**, ÉTS, Montréal, Canada, 2018
- **University Merit Financial Award**, Aligarh Muslim University, 2017–2019
- **Young Researcher Award (2nd Prize)**, HWWE Conference, 2017
- **Young Researcher Award (3rd Prize)**, HWWE Conference, 2017

Media Coverage

- **WLFI-TV News** (Sep 2026) — "Purdue Students Help Develop Underwater Robot with Multiple Ways to Move." [Link]
- **Purdue Newsroom** (Sep 2026) — "Purdue's New Underwater Robot Could Make Ocean Missions More Reliable." [Link]
- **Purdue College of Agriculture News** (Jun 2024) — "How Robots Touch on the Future of Agriculture" — coverage of tactile sensing research for agricultural robotics applications. [Link]

Presentations

- *VTAP Gripper*: Synergizing Fingertip Sensing and a Visuo-Tactile Active Palm for Dexterous In-Hand Manipulation **[IROS 2026 Best Paper Award Finalist]**
  - IEEE/RSJ IROS 2026, Pittsburgh, PA, Sep. 2026 *Poster*
- *MuxGel*: Simultaneous Dual-Modal Visuo-Tactile Sensing via Spatially Multiplexing and Deep Reconstruction
  - IEEE/RSJ IROS 2026, Pittsburgh, PA, Sep. 2026 *Poster*
- *MVP-Tac*: A Miniaturized Dual-Modal Vision and Photoelastic Tactile Sensor for Robot-Assisted Minimally Invasive Surgery
  - IEEE/RSJ IROS 2026, Pittsburgh, PA, Sep. 2026 *Poster*

- Multi-Modal Tactile Sensing for Robotic Manipulation
  - Applications & Solutions in Digital Forestry International Conference, West Lafayette, IN, Jun. 2026 *Talk*
  - Digital Forestry Mini-Symposium, West Lafayette, IN, Feb. 2026 *Talk*
  - Digital Forestry Mini-Symposium, West Lafayette, IN, Feb. 2024 *Talk*
- *VibTac*: A High-Resolution, High-Bandwidth Tactile Sensing Finger for Multi-Modal Perception
  - Midwest Robotics Workshop (MWRW), Chicago, IL, Apr. 2024 *Poster*
- *VisTac*: Towards a Unified Multi-Modal Sensing Finger for Robotic Manipulation
  - Midwest Robotics Workshop (MWRW), Chicago, IL, Apr. 2024 *Poster*
  - IROS 2023 Workshop on Visuo-Tactile Perception, Learning, and Control for Manipulation and HRI, Detroit, MI, Oct. 2023 *Poster*
  - HTM Service & Experience Innovations Symposium, West Lafayette, IN, Apr. 2023 *Poster*
  - Purdue ICON Robotics and Intelligence Systems (RISE) Expo, West Lafayette, IN, Apr. 2023 *Poster*
- Vacuum Driven Auxetic Switching Structure and Its Application on a Gripper and Quadruped
  - IEEE/RSJ IROS 2020, Las Vegas, NV, Oct. 2020 *Oral*